CYBERSECURITY PRACTICAL REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Network Walks- W2-PM-1 and W2-PM5

Pentester Name (Cybersecurity Professional)	**Sayid Ahmed Mohamed Ali**
Program/Batch	B084-Networkwalks
Date	11/11/ 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery

1. Ethical Use & Liability Disclaimer

All activities documented in this report were conducted strictly for educational and cybersecurity training purposes. Testing was limited to systems and devices that I personally own or for which I had obtained appropriate authorization.

The tools, commands, techniques, and findings presented in this report are intended to demonstrate practical cybersecurity knowledge and responsible penetration testing practices. They must not be used to access, scan, or test any system or network without proper authorization.

Unauthorized access and security testing may violate applicable laws and result in legal consequences, financial penalties, or other disciplinary actions.

I accept full responsibility for my activities and strongly support ethical hacking, responsible disclosure, and compliance with applicable cybersecurity laws and regulations.

2. Introduction

As part of my ongoing Cybersecurity Internship at **Networkwalks**, Week 2 focused on developing practical skills in two important stages of penetration testing: **Network Footprinting and Network Scanning**.

During this week's activities, I explored the information-gathering phase by performing domain footprinting on networkwalks.com using various Kali Linux tools (W2-PM1). I also conducted network scanning on my own local network using Zenmap (W2-PM5) to identify active hosts and understand the network's structure.

These practical exercises helped me understand how security professionals gather publicly available information, identify potential targets, discover active devices, and analyze network exposure during an authorized security assessment.

3. Tools and Technology Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Module W2-PM1-Reconnaissance Six Using Kali tools

During Week 2 of my Cybersecurity Internship at **Networkwalks**, I performed domain reconnaissance on networkwalks.com using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, cURL, WAFW00F, and DNSRecon**. Each tool was used to gather a different type of information about the target domain and its web infrastructure.

I began with **WHOIS** to collect publicly available domain registration details, including registrar information, registration dates, and name servers. This helped me understand the domain's registration history and identify its associated DNS infrastructure.

Next, I used **WhatWeb** to identify the technologies powering the website. The results revealed the use of **WordPress 7.1.4** and **WP Download Manager 3.3.76**, along with other web technologies. This provided an overview of the website's technology stack.

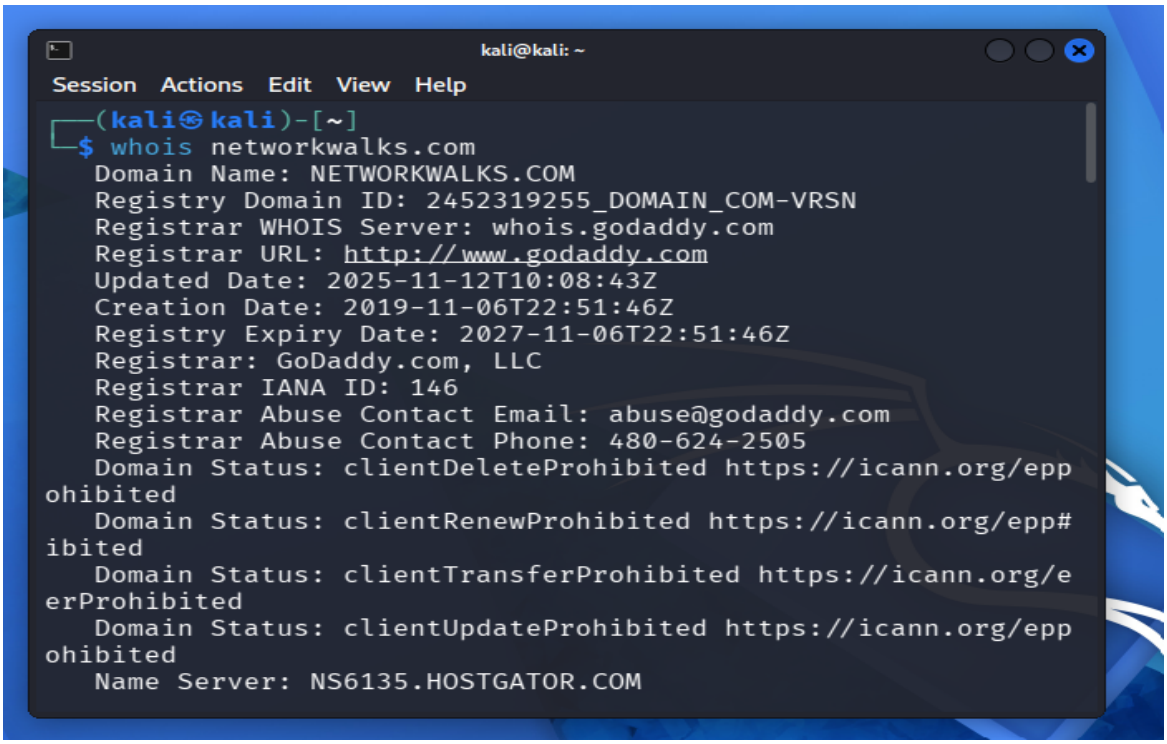
Using **Nslookup**, I resolved the domain name to its associated IP address, which was identified as **192.232.216.135**. This demonstrated how DNS translates domain names into IP addresses, allowing systems to locate web servers.

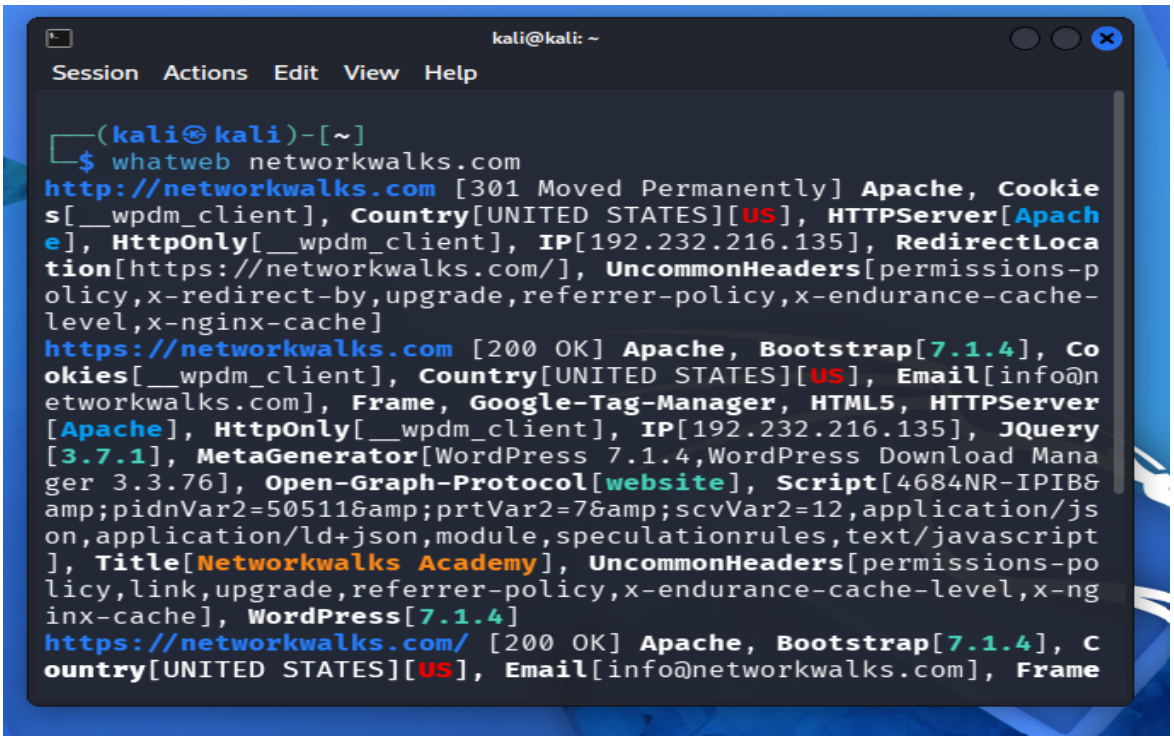
I then used **cURL** with the -I option to inspect the website's HTTP response headers and identify additional information about its web application, including the exposed WordPress REST API endpoint /wp-json/. This helped me understand how the web server communicates with clients and exposes application-related information.

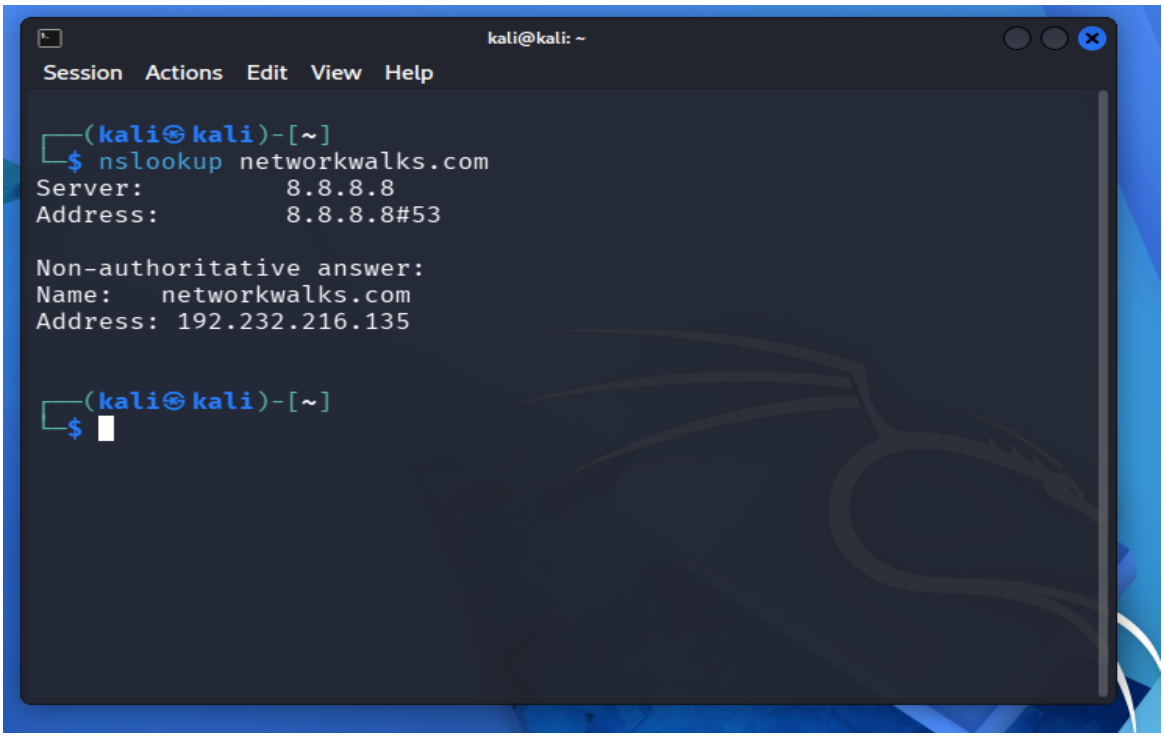
To investigate the website's security infrastructure, I used **WAFW00F** to identify the presence of a Web Application Firewall (WAF). The results indicated that **ModSecurity (SpiderLabs)** was protecting the website. This provided insight into the security technology used to help filter and monitor potentially malicious web traffic.

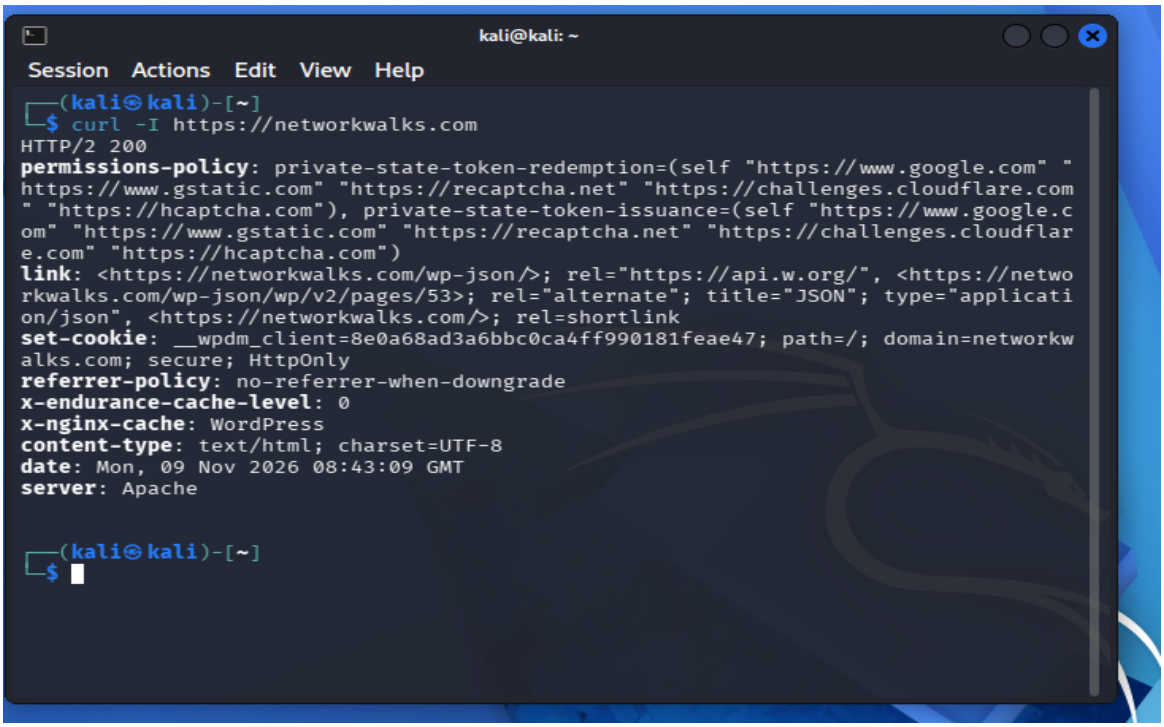
Finally, I used **DNSRecon** to perform DNS enumeration and gather additional information about the domain's DNS infrastructure, including name servers, mail servers, SPF/TXT records, service records, and other DNS-related details. This helped me understand how DNS records support domain services and infrastructure.

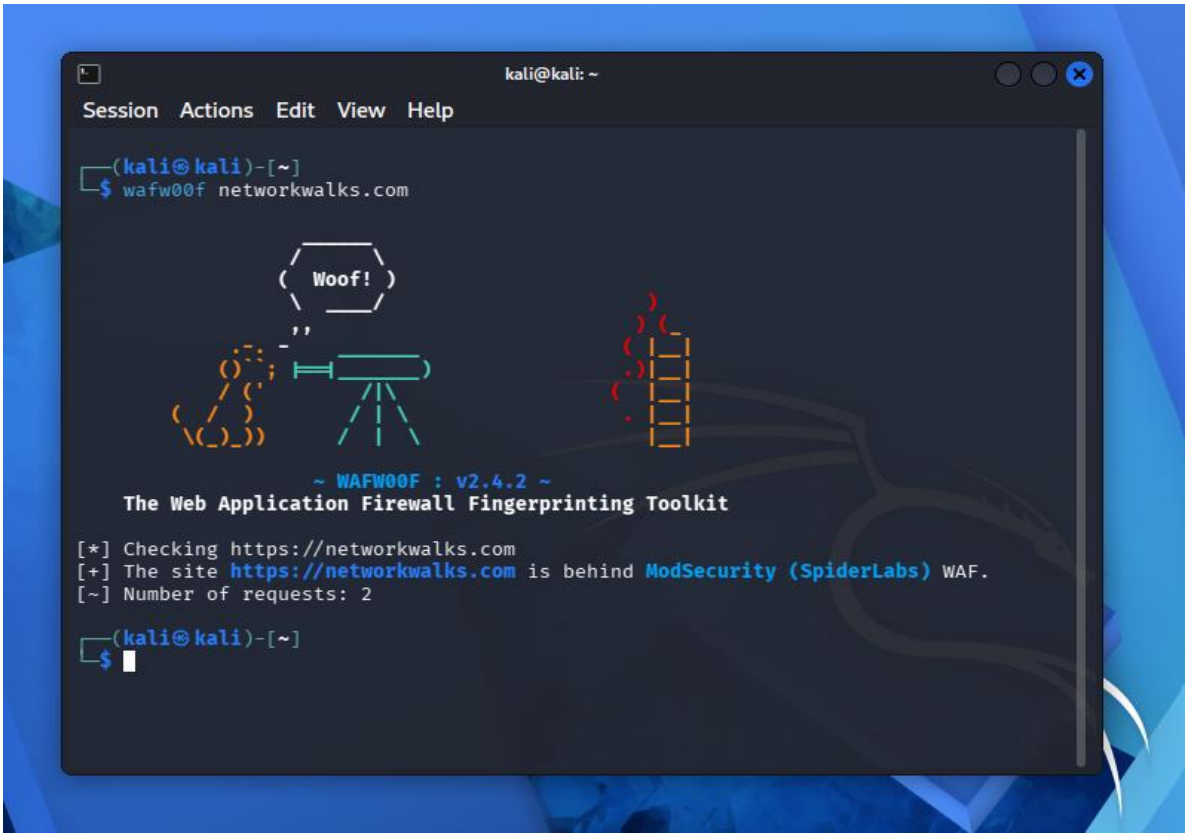
5. Evidence

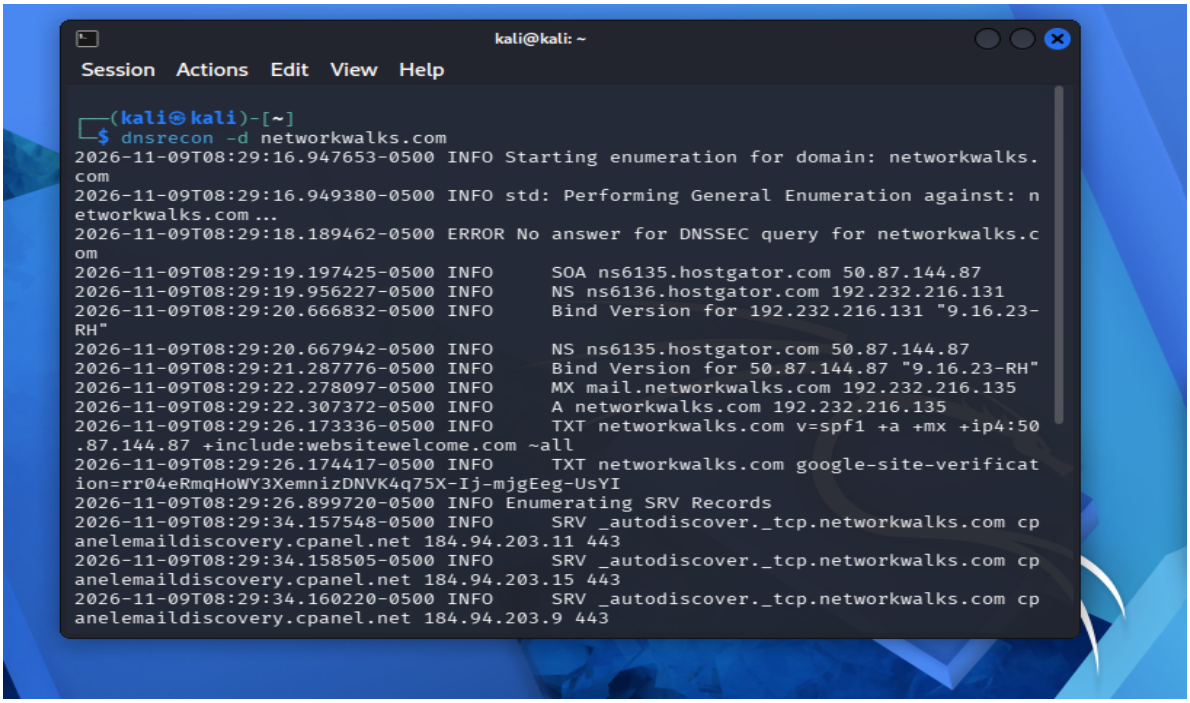

kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp
ohibited
Domain Status: clientRenewProhibited https://icann.org/epp#
ibited
Domain Status: clientTransferProhibited https://icann.org/e
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp
ohibited
Name Server: NS6135.HOSTGATOR.COM


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookie
s[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apach
e], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLoca
tion[https://networkwalks.com/], UncommonHeaders[permissions-p
olicy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-
level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Co
okies[__wpdm_client], Country[UNITED STATES][US], Email[info@n
etworkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer
[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery
[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Mana
ger 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&
amp;pidnVar2=50511&prtVar2=7&scvVar2=12,application/js
on,application/ld+json,module,speculationrules,text/javascript
], Title[Networkwalks Academy], UncommonHeaders[permissions-po
licy,link,upgrade,referrer-policy,x-endurance-cache-level,x-ng
inx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], C
ountry[UNITED STATES][US], Email[info@networkwalks.com], Frame


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nslookup networkwalks.com
Server: 8.8.8.8
Address: 8.8.8.8#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
┌──(kali㉿kali)-[~]
└─$


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=8e0a68ad3a6bbc0ca4ff990181feae47; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Mon, 09 Nov 2026 08:43:09 GMT
server: Apache
┌──(kali㉿kali)-[~]
└─$


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com
Woof!
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
┌──(kali㉿kali)-[~]
└─$


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-09T08:29:16.947653-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-09T08:29:16.949380-0500 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-11-09T08:29:18.189462-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-09T08:29:19.197425-0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-09T08:29:19.956227-0500 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-09T08:29:20.666832-0500 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-09T08:29:20.667942-0500 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-09T08:29:21.287776-0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-09T08:29:22.278097-0500 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-09T08:29:22.307372-0500 INFO A networkwalks.com 192.232.216.135
2026-11-09T08:29:26.173336-0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-09T08:29:26.174417-0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-09T08:29:26.899720-0500 INFO Enumerating SRV Records
2026-11-09T08:29:34.157548-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-09T08:29:34.158505-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-09T08:29:34.160220-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443

6. Module W2-PM5- Network Discovering Using Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover active hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local network configuration. I then entered the target subnet into Zenmap and selected the **Ping Scan** profile to identify active hosts.

The scan was performed using the following target network:

- **Target Subnet:** 192.168.254.0/24
- **Scan Profile:** Ping Scan
- **Total IP Addresses Scanned:** 256

The scan successfully identified two active hosts:

- **192.168.254.254**
- **192.168.254.1**

The scan results also identified the following MAC address:

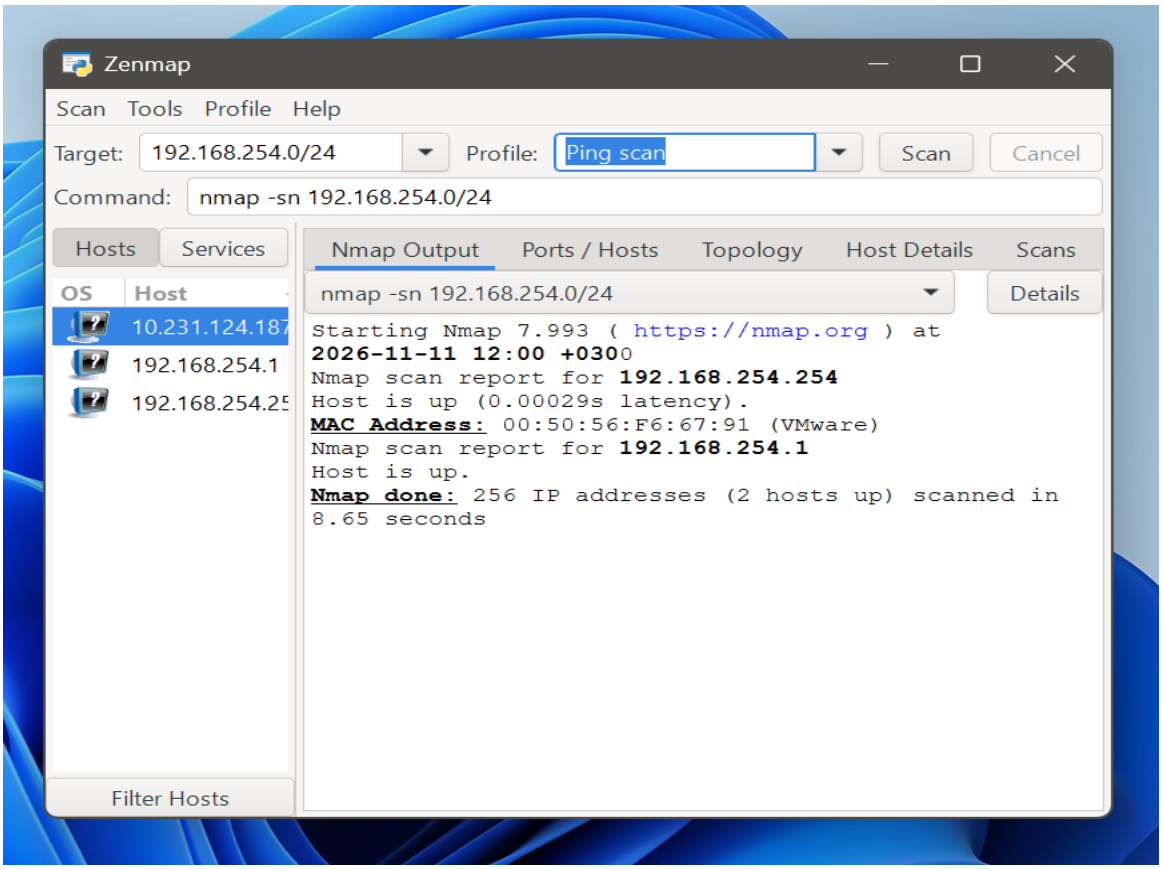
- **192.168.254.254:** 00:50:56:F6:67:91 (VMware)
- **192.168.254.1:** MAC address not displayed in the scan output.

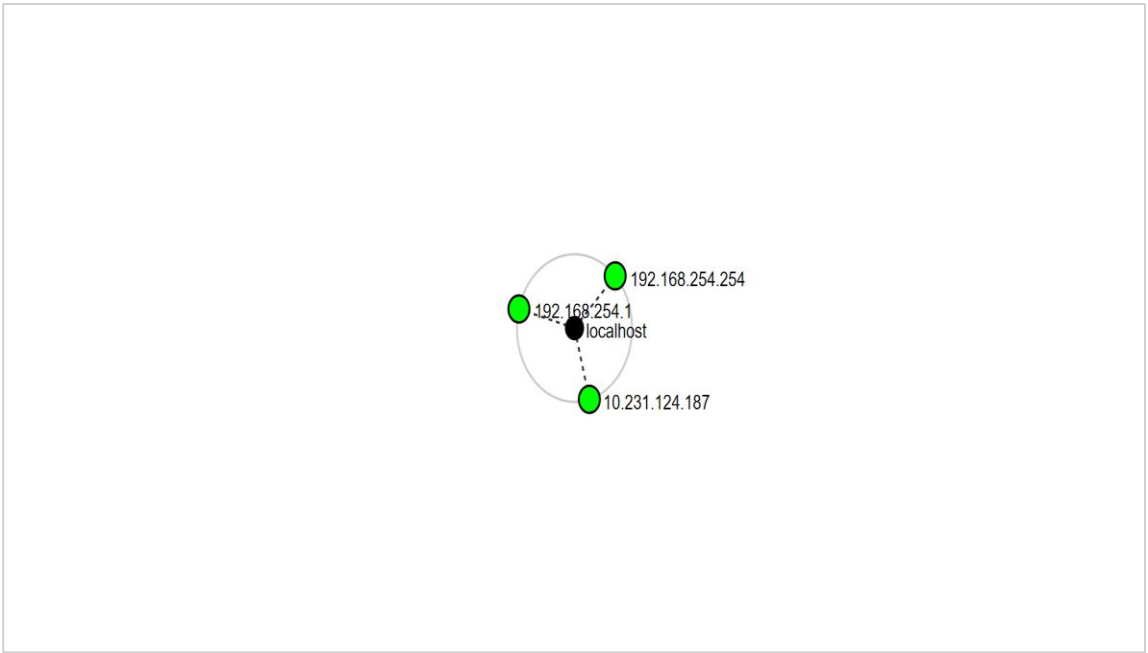
Both hosts were confirmed to be reachable. The scan completed successfully in approximately **8.65 seconds**.

After completing the scan, I opened the **Topology** section in Zenmap to visualize the discovered network devices and their connections. I enabled the topology legend and saved the network topology in PDF format as required by the practical task.

Note: The scan results documented above represent the actual results obtained from my local network during the practical exercise.

7. Evidence


Zenmap
Scan Tools Profile Help
Target: 192.168.254.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.254.0/24
Hosts
Services
OS
Host
10.231.124.187
192.168.254.1
192.168.254.25
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 192.168.254.0/24
Details
Starting Nmap 7.993 (https://nmap.org) at
2026-11-11 12:00 +0300
Nmap scan report for 192.168.254.254
Host is up (0.00029s latency).
MAC Address: 00:50:56:F6:67:91 (VMware)
Nmap scan report for 192.168.254.1
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in
8.65 seconds


192.168.254.254
192.168.254.1
localhost
10.231.124.187

8. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified two live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

9. Recommendations

Based on the findings from the reconnaissance and network scanning activities, I recommend the following security improvements to help strengthen the target website's security and improve local network visibility:

1. **Regular Software Updates:** Keep WordPress, WP Download Manager, and other website components updated to their latest stable and supported versions to reduce the risk of known security vulnerabilities.
2. **Web Technology Information Protection:** Review the website's configuration to minimize unnecessary disclosure of software versions and technical information that could assist attackers during reconnaissance.
3. **HTTP Security Configuration:** Review HTTP response headers and implement appropriate security headers to improve web application protection and reduce unnecessary information exposure.
4. **Web Application Firewall (WAF):** Maintain and regularly review the ModSecurity configuration to ensure that security rules are updated and effective in detecting and filtering potentially malicious web requests.
5. **DNS Security Review:** Regularly review DNS records, including MX, TXT, SPF, and other service-related records, to ensure they are accurate, properly configured, and do not expose unnecessary information.
6. **Network Monitoring and Device Management:** Regularly scan the local network to identify active devices and maintain an updated inventory of authorized hosts. Investigate any unfamiliar or unauthorized devices.

7. **Further Security Assessment:** Conduct additional authorized security testing, including vulnerability assessment and configuration reviews, to validate potential security weaknesses that were not examined during these practical activities.

10. Conclusion

During Week 2 of my Cybersecurity Internship at Networkwalks, I successfully completed two practical activities focusing on network reconnaissance, footprinting, and network scanning. These activities provided an opportunity to apply theoretical cybersecurity knowledge using practical tools and techniques commonly used in penetration testing.

In the first activity (W2-PM1), I used six Kali Linux tools: WHOIS, WhatWeb, Nslookup, cURL, WAFW00F, and DNSRecon to gather publicly available information about the target domain, identify its web technologies, examine DNS records, inspect HTTP response headers, and identify the Web Application Firewall. This exercise improved my understanding of how information gathering helps security professionals understand a target's digital infrastructure and potential security exposure.

In the second activity (W2-PM5), I used Zenmap to perform a Ping Scan on my local network. The scan covered 256 IP addresses within the subnet **192.168.254.0/24** and successfully identified two active hosts: **192.168.254.254** and **192.168.254.1**. I also examined the available MAC address information and generated a network topology to visualize the discovered devices.

Overall, these practical exercises strengthened my understanding of reconnaissance, DNS enumeration, web technology identification, host discovery, and network topology visualization. They also improved my confidence in using Kali Linux and Zenmap for authorized cybersecurity assessments.

Author
Sayid Ahmed Mohamed Ali
Cybersecurity Student B084

Project Information
Program Name: Cybersecurity Internship at Networkwalks | **Week:** 02 | **Repository:** GitHub